

# Chapter 3, Section 8

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1. Let  $f : X \rightarrow Y$  be a continuous map of topological spaces. Let  $\mathcal{F}$  be a sheaf of Abelian groups on  $X$ , and assume that  $R^i f_*(\mathcal{F}) = 0$  for all  $i > 0$ . Show that there are natural isomorphisms, for each  $i \geq 0$ ,

$$H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F}).$$

*Proof.* Let  $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I}^*$  be an injective resolution of  $\mathcal{F}$ . By hypothesis,  $0 \rightarrow f_*\mathcal{F} \rightarrow f_*\mathcal{I}^*$  is also exact. The pushforward of an injective object in  $\mathcal{A}b(X)$  is also injective in  $\mathcal{A}b(Y)$  because  $f_*$  is a right adjoint functor, so  $0 \rightarrow f_*\mathcal{F} \rightarrow f_*\mathcal{I}^*$  is an injective resolution of  $f_*\mathcal{F}$  in  $\mathcal{A}b(Y)$ . Suppose  $\mathcal{G} \rightarrow \mathcal{G}'$  is a monomorphism of sheaves on  $Y$ , and  $\mathcal{G} \rightarrow f_*\mathcal{I}^*$  is any morphism of sheaves. By adjunction, there is a morphism  $f^{-1}\mathcal{G} \rightarrow \mathcal{I}^*$ , and because  $f^{-1}$  is an exact functor,  $f^{-1}\mathcal{G} \rightarrow f^{-1}\mathcal{G}'$  is also a monomorphism (note that in general,  $f^*$  is only right exact). By injectivity of  $\mathcal{I}^*$ , there is an extension of  $f^{-1}\mathcal{G} \rightarrow \mathcal{I}^*$  making the following diagram commute

$$\begin{array}{ccc} & & \mathcal{I}^* \\ & \nearrow & \uparrow \\ f^{-1}\mathcal{G} & \longrightarrow & f^{-1}\mathcal{G}' \end{array}.$$

Hence, by adjunction again, there is an extension of  $\mathcal{G} \rightarrow f_*\mathcal{I}^*$  making the following diagram commute

$$\begin{array}{ccc} & & f_*\mathcal{I}^* \\ & \nearrow & \uparrow \\ \mathcal{G} & \longrightarrow & \mathcal{G}' \end{array}.$$

Since  $\mathcal{F}$  and  $f_*\mathcal{F}$  have same global sections for any  $\mathcal{F}$ , there are natural identifications

$$\begin{array}{ccccc} 0 & \longrightarrow & \Gamma(X, \mathcal{F}) & \longrightarrow & \Gamma(X, f_*\mathcal{I}^*) \\ & & \parallel & & \parallel \\ 0 & \longrightarrow & \Gamma(Y, f_*\mathcal{F}) & \longrightarrow & \Gamma(Y, f_*\mathcal{I}^*) \end{array}.$$

Taking cohomology gives the desired natural isomorphisms for all  $i \geq 0$ . □

2. Let  $f : X \rightarrow Y$  be an affine morphism of schemes (II, Ex. 5.17) with  $X$  Noetherian, and let  $\mathcal{F}$  be a quasi-coherent sheaf on  $X$ . Show that the hypotheses of (Ex. 8.1) are satisfied, and hence that  $H^i(X, \mathcal{F}) \cong H^i(Y, f_*\mathcal{F})$  for each  $i \geq 0$ .

*Proof.* Let  $V$  be an open affine subset of  $Y$ . By (8.2) and (8.5), we have

$$R^i f_*(\mathcal{F})|_V \cong H^i(\widetilde{f^{-1}V}, \mathcal{F}|_{f^{-1}(V)}),$$

which is zero because  $f^{-1}(V)$  is affine (3.7). Hence,  $R^i f_*(\mathcal{F}) = 0$ , and the result follows by the previous exercise. □

3. Let  $f : X \rightarrow Y$  be a morphism of ringed spaces, let  $\mathcal{F}$  be an  $\mathcal{O}_X$ -module, and let  $\mathcal{E}$  be a locally free  $\mathcal{O}_Y$ -module of finite rank. Prove the *projection formula* (cf. (II, Ex. 5.1))

$$R^i f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong R^i f_*(\mathcal{F}) \otimes \mathcal{E}.$$

*Proof.* The pullback of a locally free sheaf of finite rank is also locally free of finite rank. Localization is an exact monoidal functor from the category of  $\mathcal{O}_Y$ -modules to the category of Abelian groups, so by checking exactness at stalks, tensoring by  $\mathcal{E}$  is an exact functor, and it also has a left adjoint, where for any  $\mathcal{O}_Y$ -modules  $\mathcal{M}$  and  $\mathcal{N}$  (II, Ex. 5.1)

$$\begin{aligned} \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M} \otimes \mathcal{E}^\vee, \mathcal{N}) &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathbf{Hom}_{\mathcal{O}_Y}(\mathcal{E}^\vee, \mathcal{N})) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{N} \otimes (\mathcal{E}^\vee)^\vee) \\ &\cong \mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{N} \otimes \mathcal{E}). \end{aligned}$$

Hence, the result follows from the  $i = 0$  case, (1.3A), and Lemma 1.

It remains to prove

$$f_*(\mathcal{F} \otimes f^*\mathcal{E}) \cong f_*(\mathcal{F}) \otimes \mathcal{E}.$$

We first show there is a natural map

$$\phi : f_*(\mathcal{F}) \otimes \mathcal{E} \rightarrow f_*(\mathcal{F} \otimes f^*\mathcal{E}).$$

Let  $\mathcal{P}$  be the presheaf on  $Y$  defined for open subsets  $V \subseteq Y$  by

$$V \mapsto f_*(\mathcal{F})(V) \otimes \mathcal{E}(V)$$

so that  $\mathcal{P}^{\text{sh}} \cong f_*(\mathcal{F}) \otimes \mathcal{E}$ , and let  $\mathcal{Q}$  be the presheaf on  $X$  defined for open subsets  $U \subseteq X$  by

$$U \mapsto \mathcal{F}(U) \otimes f^*\mathcal{E}(U)$$

so that  $\mathcal{Q}^{\text{sh}} \cong \mathcal{F} \otimes f^*\mathcal{E}$ . Sheafification and pushforward do not commute; nonetheless there is a natural morphism  $(f_*\mathcal{Q})^{\text{sh}} \rightarrow f_*(\mathcal{Q}^{\text{sh}})$ . In particular, consider the natural morphism  $\mathcal{Q} \rightarrow \mathcal{Q}^{\text{sh}}$ . Taking the pushforward gives  $f_*\mathcal{Q} \rightarrow f_*(\mathcal{Q}^{\text{sh}})$ . Because  $f_*(\mathcal{Q}^{\text{sh}})$  is already a sheaf, by the universal property of sheafification (II, 1.2), there is unique map  $\tau : (f_*\mathcal{Q})^{\text{sh}} \rightarrow f_*(\mathcal{Q}^{\text{sh}})$  fitting into the commutative diagram

$$\begin{array}{ccc} & (f_*\mathcal{Q})^{\text{sh}} & \\ \nearrow & \text{---} \tau \text{---} \searrow & \\ f_*\mathcal{Q} & \xrightarrow{\quad} & f_*(\mathcal{Q}^{\text{sh}}) \end{array}.$$

Thus, we define a morphism of presheaves  $\psi : \mathcal{P} \rightarrow f_*\mathcal{Q}$ , which induces a morphism of sheaves  $\psi^{\text{sh}} : \mathcal{P}^{\text{sh}} \rightarrow (f_*\mathcal{Q})^{\text{sh}}$ , and composing with  $\tau$  gives our desired morphism

$$\phi = \tau \circ \psi^{\text{sh}} : \mathcal{P}^{\text{sh}} \rightarrow (f_*\mathcal{Q})^{\text{sh}} \rightarrow f_*(\mathcal{Q}^{\text{sh}}).$$

The pushforward for the presheaf  $\mathcal{Q}$  distributes over the tensor product, that is

$$\Gamma(V, f_*\mathcal{Q}) = f_*(\mathcal{F})(V) \otimes f_*f^*(\mathcal{E})(V).$$

There is a natural morphism of sheaves  $\iota : \mathcal{E} \rightarrow f_*f^*(\mathcal{E})$ , so take  $\psi = \text{id}_{f_*\mathcal{F}} \otimes \iota$  to be our morphism of presheaves

$$\begin{array}{ccccc} & & \phi & & \\ & \searrow \psi^{\text{sh}} & \nearrow & \searrow \tau & \\ \mathcal{P}^{\text{sh}} & \xrightarrow{\quad} & (f_*\mathcal{Q})^{\text{sh}} & \xrightarrow{\quad} & f_*(\mathcal{Q}^{\text{sh}}) \\ \uparrow & & \uparrow & \nearrow & \\ \mathcal{P} & \xrightarrow{\quad \psi \quad} & f_*\mathcal{Q} & & \end{array}.$$

Now that we defined a morphism, we remark that the question is local on  $Y$ . In particular, let  $y$  be a point in  $Y$ , and let  $V$  be an open subset of  $Y$  such that  $\mathcal{E}$  is free of finite rank. Then  $f^*\mathcal{E}$  is free of finite rank on  $U = f^{-1}(V)$ . Consider the commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\bar{f}} & V \\ i \downarrow & & \downarrow j \\ X & \xrightarrow{f} & Y \end{array}.$$

Because  $i$  and  $j$  are open immersions, there is an isomorphism of functors

$$j^*f_* \cong \bar{f}_*i^*, \quad i^*f^* \cong \bar{f}^*j^*.$$

Pullback distributes over tensor products, so we have

$$\begin{aligned} j^*f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong \bar{f}_*i^*(\mathcal{F} \otimes f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes i^*f^*\mathcal{E}) \\ &\cong \bar{f}_*(i^*\mathcal{F} \otimes \bar{f}^*j^*\mathcal{E}), \\ j^*(f_*(\mathcal{F}) \otimes \mathcal{E}) &\cong j^*f_*(\mathcal{F}) \otimes j^*\mathcal{E} \\ &\cong \bar{f}_*(i^*\mathcal{F}) \otimes j^*\mathcal{E}. \end{aligned}$$

Open immersions preserve stalks, so we can assume  $\mathcal{E}$  to be a free  $\mathcal{O}_X$ -module of finite rank  $n$ . The pullback of a free sheaf of finite rank is also free of finite rank, so we have the isomorphisms

$$\begin{aligned} f_*(\mathcal{F} \otimes f^*\mathcal{E}) &\cong f_*(\mathcal{F} \otimes \mathcal{O}_X^{\oplus n}) \\ &\cong f_*(\mathcal{F}^{\oplus n}) \\ &\cong f_*(\mathcal{F})^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{O}_Y^{\oplus n} \\ &\cong f_*(\mathcal{F}) \otimes \mathcal{E}. \end{aligned}$$

The third isomorphism holds because  $f_*$  is right adjoint, and right adjoint functors preserve limits, and a direct sum of finite number of terms can be realized as a limit, so the pushforward and finite direct sums commute.  $\square$

**Lemma 1.** Let  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  be Abelian categories, and assume  $\mathcal{A}$  and  $\mathcal{B}$  have enough injectives. Let  $F : \mathcal{A} \rightarrow \mathcal{B}$  and  $G : \mathcal{B} \rightarrow \mathcal{C}$  be functors such that  $F$  has a left adjoint (or more generally preserves injectives), and either

- (a)  $F$  is left exact, and  $G$  is exact; or
- (b)  $F$  is exact, and  $G$  is left exact.

In each case, we have the following natural isomorphisms

- (a)  $R^i(G \circ F) \cong G \circ R^i(F)$ ;
- (b)  $R^i(G \circ F) \cong R^i(G) \circ F$ .

*Proof.* The proof for (b) is similar to the proof of Exercise 1, so we prove (a). Let  $A$  be an object in  $\mathcal{A}$  and choose an injective resolution

$$0 \rightarrow A \rightarrow I^*,$$

where one has

$$R^i F(A) = h^i(FI^*).$$

Applying  $G \circ F$  gives a complex in  $\mathcal{C}$

$$0 \rightarrow (G \circ F)A \rightarrow (G \circ F)I^*,$$

and we want to show

$$h^i((G \circ F)I^*) \cong G \circ h^i(FI^*).$$

But this follows from exact functors commuting with cohomology.

□